

Completeness $\wedge$ Transitivity $\Rightarrow$ Weak order

Result — human; produced by definition.

0.1 Premises

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.

0.2 Conclusion

- **Weak order.** $\succcurlyeq$ is complete and transitive.

0.3 Proof

By definition of weak order.

Record: topics/decision-theory/results/weak-order-def-forward.yaml